

Swagotam Malakar

Computer Science & Engineering Student

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Education

- 2022–2026 **B.Sc. in Computer Science & Engineering**, United International University, Dhaka, Bangladesh. (Expected)
- 2020 **Higher Secondary Certificate (HSC)**, Govt. Science College, Dhaka, Bangladesh, GPA: 5.00/5.00.
- 2018 **Secondary School Certificate (SSC)**, Jury Govt. Model High School, Sylhet, Bangladesh, GPA: 5.00/5.00.

Undergraduate Thesis (FYDP)

- Jan 2025–Present **Multi-Dimensional Center of Mass Estimation**, for Uniform and Non-Uniform Objects. Supervised Research (Thesis)
- Goal:** To develop a non-contact method for estimating the Center of Mass (CoM) of irregular, non-uniform objects using Computer Vision and Machine Learning, where classical physical balancing methods fail.
- Technical Achievements:**
- **3D Reconstruction:** Developing a pipeline to generate 3D mesh representations of physical objects from 2D camera feeds using photogrammetry techniques.
 - **Physics-Informed ML:** Implementing **PointNet++** for semantic segmentation of 3D point clouds to identify density variations in non-uniform objects.
 - **Data Pipeline:** Designing a robust sensor data processing workflow to handle high-dimensional spatial data for real-time estimation.

Independent Research

- Nov 2025–Present **Privacy-Preserving Invasive Ductal Carcinoma (IDC) Detection**, via Federated Learning with GAN Augmentation. Research Project (Submission Pending)
- Goal:** To develop a decentralized framework for detecting breast cancer (IDC) patches from histopathology images without centralized data sharing.
- Technical Achievements:**
- **Non-IID Handling:** Implemented **FedProx** and **FedAvg** algorithms to handle statistical heterogeneity across simulated hospital clients.
 - **GAN Augmentation:** Integrated **DCGAN** to generate synthetic minority class samples locally, resolving severe class imbalance (IDC-negative vs. positive).
 - **Model Optimization:** Utilized **EfficientNet-B0** with Focal Loss to maximize feature extraction efficiency on edge constraints.
 - **Results:** Achieved ~ 0.86 AUC with improved calibration ($ECE \approx 0.37$) on the global test set.

- Dec 2025–Present **Bornomala, Bangla Dialect Normalization Pipeline.**
Research Project
- Goal:** To bridge the performance gap of ASR systems on regional dialects (Sylheti, Chittagonian) by solving the "Language Identity Crisis" through domain-specific data acquisition.
- Technical Achievements:**
- **Data Mining Pipeline:** Developed an automated scraper to harvest unstructured dialect audio from **YouTube** (dramas/vlogs), augmenting the failure-prone RegSpeech12 dataset.
 - **Smart Segmentation:** Integrated **Silero VAD** to intelligently remove silence and background noise, creating high-fidelity speech segments for training.
 - **Transcription:** Deployed **Whisper (Medium)** for robust automated draft transcription, utilizing aggressive memory management for optimization on P100 GPUs.
- Aug 2025–Jan 2026 **Swarabyanjan, Yellow Journalism Detection via Stance Detection.**
Manuscript in Preparation
- Goal:** To automatically detect sensationalism and "clickbait" in Bengali news by verifying the semantic consistency between headlines and article bodies.
- Achievements So Far:**
- **Model Architecture:** Fine-tuned **BanglaBERT (Cross-Encoder)**, achieving SOTA performance by leveraging deep bidirectional context for NLI (Natural Language Inference).
 - **Hard Negative Mining:** Utilized **BM25** to retrieve "Hard Negatives" – article bodies that are lexically similar to headlines but factually unrelated – forcing the model to learn deep semantic logic rather than keyword matching.
 - **Logic Injection:** Created a targeted synthetic dataset to resolve **"Fictional vs. Real"** ambiguities (e.g., distinguishing "Actor dies in movie" from real news), eliminating false positives in entertainment news.

Major Projects

- Nov 2025–Feb 2026 **Cyber Aware Prohory, Machine Learning Developer.**
- Overview:** An automated classification system built to identify and filter text-based threats.
- Key Features:**
- **Threat Detection Pipeline:** Developed and trained models to accurately categorize text inputs as spam, danger, or safe messages.
 - **Model Optimization:** Conducted iterative testing and evaluation of progressive model architectures (including versions v26, v27, and v28) across diverse text inputs to maximize classification performance.
- Jun 2025–Present **AmarShohor – Smart City Platform, Team Lead.**
- Overview:** A platform integrating civic engagement, real-time services, and utility connectivity.
- Key Features:**
- **Civic Services:** Issue reporting system, blood donation matching, and NGO/Utility connectivity.
 - **Smart Routing:** Implemented SOS alerts and dynamic map formulation using ML-powered priority management.
- Tech Stack:** React Native, Node.js, Express.js, Django (ML), MongoDB, PostgreSQL.
- Feb 2025–Jun 2025 **LifeLink – All-in-One Service Platform, Full-Stack Developer.**
- Overview:** Unified digital platform for healthcare, education, and employment services.
- Key Features:**
- Designed a custom MVC architecture supporting 8 service modules with 156 API endpoints.
 - Integrated secure payment gateways (bKash/Nagad) and an AI-powered chatbot.
 - Optimized database queries to support 10,000+ concurrent users with sub-2.3s load times.
- Tech Stack:** PHP 7.4+, MySQL, JavaScript, Tailwind CSS.

Oct **Blockchain Supply Chain**, *Blockchain Developer*.
2024–Jan **Overview:** Ethereum-based solution to combat counterfeit drugs in the pharmaceutical supply chain.
2025 **Tech Stack:** Solidity, Web3.js, React.js, Ethereum, Truffle.

Technical Skills

Languages Python, C/C++, Java, JavaScript (ES6+), PHP, Solidity, LaTeX
AI & Federated Learning (FedProx), BanglaBERT, Whisper, Silero VAD, BM25, GANs, PyTorch
Research
Web & Tools React.js, Node.js, Django, Docker, AWS, Git, CI/CD, Linux

Certifications

Development Stanford CS106A (Code in Place), HackerRank (Software Engineer), Cisco CLA (C Programming), HackerRank (Python Basic), Udemy (Java Programming), Udemy (LeetCode in Java)
Networking CCNA: Introduction to Networks, Cybersecurity Essentials (Cisco), NDG Linux Essentials

Honours & Awards

Academic Board Scholarship (PSC, JSC, SSC), Perfect GPA (5.00) in all public exams
Competitions Multiple awards in National Science Olympiads and Inter-College Quiz Championships

Extracurricular Activities

Communities **UIU Competitive Programming Community** - Active Member (2025-Present)
Social Service **Bangladesh Scouts** - Active member (Jan 2010 - May 2018). Participated in community service projects, disaster relief operations, and environmental conservation.
Creative **Bengali Novelist** - Authoring Bengali fiction exploring social philosophy and human identity.

Language Proficiency

English	Professional Working Proficiency	<i>Written: Excellent / Spoken: Fluent / Reading: Excellent</i>
Bangla	Native Speaker	<i>Mother tongue</i>
Hindi	Conversational	<i>Written: Poor / Spoken: Good / Reading: Basic</i>

Interests

Tech AGI, Machine Learning (ML), Deep Learning (DL), Computer Vision, Blockchain, Cybersecurity
Research Natural Language Processing, Bengali Language Computing, Data Mining, AI Ethics
Others Astrophysics, Reading novels, Philosophy of science, History

References

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